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The relation between lateral thinking and inquiry skills of higher education students: A path analysis

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Abstract

The aim of the study is to examine the relations between higher education students' lateral thinking dispositions and inquiry skills. Survey design and Structural Equation Modelling were used in the study. The sampling of the study consisted of 975 prospective teachers attending three faculties of education in three different state universities in Turkey. Inquiry Skills Scale and Lateral Thinking Disposition Scale were used to collect data. The collected data were analysed using R Studio programme. It was observed in the study that students' inquiry skills and lateral thinking dispositions were high. It was revealed in the study that inquiry skills predicted lateral thinking dispositions. Inquiry skills account for 31% of lateral thinking disposition. Thus, it can be concluded that lateral thinking dispositions of higher education students' depend on their inquiry skills at a percentage of 31%.

KEYWORDS
higher education, inquiry, path analysis, relation, thinking

1 | INTRODUCTION

At present, designing and performing the instructional activities are closely related to the qualities of the individuals needed in a particular society. In the 21st century, societies need individuals who question, inquire, come up with creative solutions to problems and who have higher order thinking skills ranging from critical thinking

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to creative thinking. Thus, it can be stated that the curricula are expected to pave the way for students to think through developing the thinking skills of individuals (Kurnaz, 2013, pp. 1–2). Thinking is an indispensable act in everyday life for all people. Human beings use the act of thinking intensively for many different purposes such as problem solving, understanding, building links, foreseeing, decision making, selection and comparison (Semerci et al., 2019). The way we think affects our plans, objectives and decisions. It seems difficult to make a comprehensive definition of thinking as it is an abstract concept and has many dimensions. Therefore, it is not unusual to see differing definitions of thinking in literature. Halpern (1997, p. 3) describes thinking as "the process of creating new knowledge through prior knowledge". Baron (2008) defines thinking as a high-level cognitive function used in the processes of making choices and judgements. It is a mental activity used to eliminate suspects on what to do, what to believe or what to pursue (Baron, 1993). Finally, Mayer (1983) indicates that thinking is a cognitive, goal-directed process and it can be observed in behaviour. Thus, the common characteristics of thinking observed in the above definitions are that it is a cognitive activity, it is used to make choices and create new knowledge and it is goal-directed.

In educational contexts, thinking is usually defined as "a consciously goal-directed process, such as remembering, forming concepts, planning what to do and say, imagining situations, reasoning, solving problems, considering opinions, making decisions and judgments and generating new perspectives" (Moseley et al., 2005, p. 12). A great deal of research suggests that thinking should be taught in school (Brewerton, 2010; Burden, 1998; Csapo, 2001; Scheinin & Mehtalainen, 1999) as it is an integral part of learning. The need to teach thinking skills in schools has arisen due to societal demands, requirements of employment and sophisticated nature of everyday life (McTighe & Schollenberger, 1991). As Moseley et al. (2005, p. 9) emphasise, "thinking skills are needed, not only in the worlds of work, education and training but in the contexts of family, friendship and community and in the construction of personal and shared beliefs and values". Also, students have been inevitably involved in the digital learning contexts for the past two years as a result of COVID-19 pandemic, and digital learning has compelled students abruptly to develop skills of different types of thinking (Qi, 2021) that will be used fruitfully in the years to come. While giving the justifications for incorporating thinking skills, especially creative and lateral thinking, into educational processes, Kumari and Aggarwal (Kumari & Aggarwal, 2012, p. 32) indicate that "education system is so information oriented that it gives ready-made answers killing the natural tendency of students to explore, experiment and to experience". They also complain that schools mainly focus on verbal, rational and time-based left hemisphere of the brain, neglecting the other. In addition, El Soufi and See (2019) indicate in their study that higher education students have lacked higher-order thinking skills especially for the past thirty years. However, research evidence demonstrates that attempts have been launched to foster higher-order thinking skills of higher education students through technology-based applications (Lu et al., 2021; Qi, 2021; Rababa, 2021; Varenina et al., 2021; Yulian, 2021).

Thinking skills are studied under different sub-categories and some well-known ones are creative thinking, critical thinking, reflective thinking, analytical thinking, convergent thinking, and divergent thinking. De Bono (1990) asserts that conventional thinking methods enable us to create fixed patterns of concepts and to improve and validate those patterns. However, we need more than conventional methods of thinking to create new patterns and restructure the existing ones, which frees us from the prevalence of outdated patterns of thinking. At this point, the term *lateral thinking* comes to the fore. Lateral thinking was originated by Edward de Bono in late 1960s to supplement creative thinking. It was put forward as a reaction to tightly structured or "standardised" way of thinking, which didn't allow for different or alternative perspectives.

Edward de Bono is regarded by many as the leading international authority on teaching thinking, especially creative thinking. Lateral thinking can be considered as the extension of thinking skills by adding a new dimension. Waks (1997) regards the emergence of lateral thinking as a natural response to the "bombardment" of people in recent years with enormous amount of information. De Bono (1990) defines lateral thinking as skipping one pattern to another in an asymmetric model creatiDe oon system. For Lawrence and Xavier (2013, p. 29) lateral thinking is "a way of thinking which seeks the solution to intractable problems through unusual methods or elements which

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would normally be ignored by logical thinking". While studying thinking skills, De Bono's main aim was to find the "sideways" to think creatively and to cease the judgmental or adversarial nature of habitual thinking patterns (Burgh, 2014).

At this point, the phrase *habitual thinking patterns* mainly refers to two terms: vertical thinking, as de Bono calls it, and critical thinking. De Bono criticised sequential, logical or linear thinking, which he later called vertical thinking, along with critical thinking as he believed they would impede or hinder potential creative abilities of individuals. De Bono coined the term lateral thinking as a complementary equivalent of vertical thinking; and parallel thinking, which is cooperative and coordinated thinking, to replace critical thinking. Lateral thinking is needed when ordinary vertical thinking is unable to find a solution to a problem or when a new idea is required (De Bono, 2014) and it aims at "freeing the mind from the imprisonment caused by already established concepts and patterns" (Waks, 1997, p. 245). Chang and Zhou (2022) highlight this characteristic of the term stressing that lateral thinking depends on the indicators of imagination, perception and attention. With vertical thinking, one is confined to follow already-paved roads to go from A to B and it is almost impossible to find alternative routes or shortcuts; however, lateral thinking enables people to think alternatively and gain insight in the process of thinking. Thus, in addition to finding alternative ways to destination B, one can even change the destination as a result of lateral thinking. Therefore, it can be claimed that lateral thinking rejects any kind of boundaries or frameworks. Vertical thinking is a predictable and high-probability thinking that is employed by a large number of people and it doesn't usually allow for the individual differences to emerge as much as lateral thinking does. However, lateral thinking enables people to exhibit their individual styles of thinking and problem-solving approaches. Thus, lateral thinking can be described as a way of thinking which requires breaking the habitual patterns of thinking, which were essentially employed by vertical thinking and eventually paves the way for creative thinking.

De Bono does not make a distinction between vertical and lateral thinking with the aim of demonstrating which one is superior or inferior to the other. Rather, he emphasises the differences to use both terms effectively as both of them are necessary to think and produce alternative ways. He regards vertical thinking as the conventional way of thinking that everybody employs, whereas he considers lateral thinking as a way of thinking that aims at creating new patterns and restructuring the old ones to put forth mostly rare, unexpected or unusual thoughts. As De Bono (2014) maintains, lateral thinking is not a new, magic formula but simply a different and more creative way of using the mind. While using these two thinking skills for problem-solving, lateral thinking should precede traditional methods of vertical or logical thinking to have an effective output (Burgh, 2014) as confirmed by De Bono (1979, p. 12) stating that "there is no harm in rationalizing a vertical-thinking path to the solution after it has been reached by lateral thinking".

Lateral thinking is closely related to insight, creativity and humour (De Bono, 1977). These three constructs are usually God-given, whereas lateral thinking can be acquired later. As De Bono (1971, p. 13) indicates "it involves open-ended scanning techniques for generating options in the formative phases prior to decision selection, implementation and assessment". What is more, "the skill of lateral thinking gives a person the opportunity to generate new ideas in the creative process" (Qi, 2021, p. 2). Lateral thinking could be used deliberately in cases like generating new ideas, problem solving, processing perceptual choice, periodic reassessment of assumptions, and prevention of sharp divisions and polarizations (De Bono, 1977; Elaldı, 2021). Thus, it can be asserted that lateral thinking is quite useful in urgent or troublesome cases.

Having a clear idea about the theory of lateral thinking is not enough on its own as it is essential to put theory into practice to make it more substantial. In the related literature, it is frequently stressed to prefer lateral thinking over vertical thinking to make a difference as vertical thinking is used by lots of people and produces expected or predictable results. In order to arrange the ways used during lateral thinking for a better and more practical use of this type of thinking, some techniques have been introduced by de Bono. Major techniques are; generating alternatives, challenging assumptions, suspended judgement, fractionation, reversal method, brainstorming, analogies, division, polarisation and random stimulation. One shouldn't be confined to these "prescribed" techniques

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while thinking laterally. De Bono (1977) states that the purpose of these techniques is to provide an opportunity for the practical use of lateral thinking so that one can gradually acquire the lateral thinking habit. Once the lateral thinking habit is acquired, some personal techniques can also be used as long as the principles of lateral thinking are followed.

Researchers studied lateral thinking for different purposes. While some researchers (Karagöz, 2019; Lawrence & Xavier, 2013; Semerci, 2017; Yıldız & Yılmaz, 2020) just aimed at identifying the lateral thinking levels of higher education students with respect to some variables such as gender, department and seniority, some others studied the effect of lateral thinking skill on the development of English speaking skills (Nageswari et al., 2016), academic achievement (Prakash & Vasimalairaja, 2021), creative thinking (Srikongchan et al., 2021), quality of instruction in higher education (Selvam & Arasu, 2021; Ślaski & Grzelak, 2022), development of turnaround leadership and sustainable development goals of UN (Ghasemy et al., 2023) and group work (Elaldı, 2021). In addition, a group of researchers studied the relation between lateral thinking with intelligence and achievement (Kumari & Aggarwal, 2012), technology education (Waks, 1997), speaking competence (Omran & Krebet, 2022), social innovation (Aydemir, 2021), emphatical tendency (Arslan et al., 2022) and problem solving skills (Yazgan, 2021). More specifically, studies on mathematical lateral thinking were also observed (Julita & Herman, 2019; Nggaba et al., 2018; Susilawati et al., 2018). However, it could be claimed that the quantitative and qualitative research on lateral thinking conducted so far fails to appeal to the relation(s) lateral thinking has with inquiry (skills), which necessitates further relational research.

Along with lateral thinking, the second variable focused on in this study is inquiry skills. Thinking and inquiry are the heart of cognitive skills (Güneş, 2012) and inquiry is essential to think (Yılmaz, 2019). Inquiry is a term in education and daily life that denotes seeking explanation or information by asking questions (Harlen, 2013). In addition, Martin-Hansen (2002, 35) points out that "inquiry includes the activities of students—such as posing questions, planning investigations, and reviewing what is already known in light of experimental evidence—that mirror what scientists do". Moreover, inquiry requires using critical as well as logical thinking, considering alternative explanations, and identifying assumptions (National Research Council, 1996). In essence, "inquiry is an active learning process in which students answer research questions through data analysis" (Bell et al., 2005).

Inquiry skills, on the other hand, can be described as the skills which are required to dig beneath the existing information or knowledge and which are one of the main sources for the implementation of other 21st century skills (Aslan, 2017). Inquiry skills are important in that they play a significant role in student learning and adapting to social life and life processes (Lim, 2001). It is noteworthy that higher-order mental processes are used in establishing an experience, individuals' discovering themselves, being curios and observant, and focusing on problem-solving while acquiring inquiry skills (Dow, 2000, p. 5). What is more, inquiry skills demand that students engage actively in collecting information, analysing findings and employing critical thinking concerning what is learned. Finally, using inquiry skills in the process of learning arouses curiosity, improves critical thinking, enhances student responsibility, encourages independent thinking and promotes reasoning skills (Honzel, 2012).

A great deal of research indicates that research on inquiry learning mainly focuses on inquiry skills (Bailey, 2018; Bell et al., 2005; Gormally et al., 2009; Lim, 2001; Pedaste et al., 2015; Tornee et al., 2017). It is noteworthy that higher-order mental processes are employed in individuals' building an experience, exploring themselves, observing, and focusing on problem solving (Dow, 2000, p. 5). Inquiry learning is usually defined as a process of exploring and it requires using a collection of skills that are essential in any aspect of life. Thus, inquiry skills are crucial with respect to adaptation to social life and vital processes (Lim, 2001). Individuals having inquiry thinking skills are able to employ scientific thinking processes and improve themselves in building knowledge through prioritising research and getting information (Lin, 2007). Inquiry learning helps individuals in the process of critical thinking and problem-solving in educational settings by encouraging questioning and checking the accuracy of newly acquired information. In addition, it is considered that individuals could acquire awareness by being equipped with

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the possible solutions of the problems they can encounter in their lives with the contribution of inquiry thinking strategies in the context of education (Costa, 1991). In this respect, it is quite important to incorporate inquiry skills more in curriculum studies with regards to supporting individuals' academic achievement as well as their cultural development. Therefore, integrating learning based on inquiry skills is necessary while identifying the syllabi of the courses (Bailey, 2018, p. 11).

In the educational contexts based on inquiry thinking, the teacher guides students in the process and supports student learning based on experience with various case studies and explore themselves in learning-teaching process (Costa, 1991). In order to develop inquiry skills of students, it is necessary to evaluate the level of the skill in question in the supervision of teacher and to make use of different models in different learning contexts (Davies et al., 2012). In the instructional processes where inquiry-based learning environments are encouraged, the necessity of identifying the shortcomings of instructors in inquiry skills and improving them is usually emphasised as the instructors are guides and leaders who play an active role in inquiry learning (Lim, 2001). In inquiry learning environments, the guidance of teachers contributes to students' learning through inquiry thinking taking into account new ideas (Harrison, 2014). It is crucial for teachers to give priority to real life implementations such as nature-environment education which enable students to be involved in studies with a critical viewpoint and to explore themselves while guiding students in learning environments that promote inquiry skills (Tornee et al., 2017). Given the important role of teachers in improving inquiry skills of students, their competence regarding inquiry skills during preservice training period gain greater importance. Thus, the higher the level of inquiry skills of teachers and prospective teachers are, the higher their competence in teaching inquiry skills to their students will be.

When the related literature was reviewed, it was observed that prospective teachers' inquiry skills were examined in some studies (Abalı Öztürk et al., 2017; Aldan Karademir & Saracaloğlu, 2017; Bedir & Duman, 2017; Dolapçıoğlu, 2020; Hayat & Rustaman, 2017; Katwijk et al., 2021; Şahin et al., 2017; Saputra et al., 2019; Yıldız Feyzioğlu, 2015). However, it was found out that the majority of the studies sought the relation between inquiry skills and critical thinking skills, especially the effect of inquiry based learning on critical thinking (Al-afifi & Ambusaidi, 2014; Damopolii et al., 2021; Duran & Dökme, 2016; Ghaemi & Ghazi, 2017; Madhuri et al., 2012; Magnussen et al., 2000; Qing et al., 2010; Seranica et al., 2018; Wale & Bishaw, 2020). It was discovered that the relation between lateral thinking and inquiry skills of prospective teachers or of any other population has not been studied before. This was the main inspiration of the researchers involved in this study. Inquiry is related with critical thinking and lateral thinking is related with creative thinking. Thus, this study could be characterised as a study which investigates the relations between critical and creative thinking indirectly. Creativity and critical thinking are listed as the two main 21st century skills under the category *learning and innovation skills* (Partnership for 21st Century Skills, 2009) and 21st century skills are interconnected (Alkhayari, 2021; Myers, 2019). So, it is a good idea to study the relationship between lateral thinking, related with creativity, and inquiry, related with critical thinking.

2 | PURPOSE OF THE STUDY

This study examines the relationship between higher education students' inquiry skills and lateral thinking dispositions. To this end, answers for the following questions are sought:

1. What is the level of higher education students' inquiry skills and lateral thinking dispositions?
2. Do higher education students' inquiry skills and lateral thinking dispositions differ with respect to the department they are enrolled in and their class levels?
3. Do inquiry skills predict lateral thinking?

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2.1 | Method

2.1.1 | Research design

In the first stage, survey design was used to collect data. Data on higher education students' lateral thinking dispositions and inquiry skills were collected through two scales and analysed using descriptive statistical methods. Creswell (2003) indicates that survey design provides researchers with a quantitative or numeric description of trends, attitudes, or opinions of a population, enabling them to make generalisations or draw inferences to the population. In the second stage, Structural Equation Modelling was employed in the study which "combines multiple regression, path analysis and factor analysis" (Fraenkel et al., 2012, p. 338). As MacLean and Gray (1998) point out "Structural Equation Modeling provides a statistically valid means of using the information we obtain through measurement to calibrate the relationships we hypothesise to exist between the underlying (latent) non-measurable variables". What is more, it is one of the means that are employed in social sciences to evaluate the relations between or among variables and to test the theoretical models (Joreskog & Sörbom, 2001). A path analysis was conducted as the three sub-dimensions of inquiry skills were included in the correlational analysis in addition to main variables, lateral thinking disposition and inquiry skills. For Fraenkel, Wallen and Hyun (Fraenkel et al., 2012, p. 337), "path analysis is used to test the likelihood of a causal connection among three or more variables". Thus, a comprehensive analysis was carried out concerning the relations among the variables in the study.

2.1.2 | Study group

The study group consisted of 975 higher education students attending the faculties of education of three state university in Turkey. The demographic data concerning the participants are given in Table 1.

As seen in Table 1, 724 (74.3%) of the participants are female, whereas only 251 (25.7%) are male. That means the number of female participants almost triples that of male ones as most of the students at the faculties of education in Turkey consist of female students. It can be claimed that there is a balanced distribution of participants in

TABLE 1 Demographic data on the participants of the study.

	f	%
Gender		
Female	724	74.3
Male	251	25.7
Department		
Turkish Language Education	267	27.4
Science Education	85	8.7
Mathematics Education	162	16.6
Elementary Education	302	31.0
Social Sciences Education	159	16.3
Class level		
1st year	226	23.2
2nd year	391	40.1
3rd year	194	19.9
4th year	164	16.8
Sum	975	100

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terms of the departments they were enrolled in as the number of students for each department ranges between 159 and 362 except for those attending the department of science teaching ($f = 85$), which can be accounted for comparatively lower number of students in that department. As for the class levels of students, the number of students at each class level ranges from 164 to 391, fourth year students being the lowest in number and second year students are highest.

2.1.3 | Data collection tools

Lateral thinking disposition scale

The Lateral Thinking Disposition Scale was developed by Semerci (2016). It is a five-point Likert scale and consists of nine items and just one dimension. The KMO value of the scale was 0.794 and Barlett's test value was 1,585,363 (Df = 36, $p = .000$). The scale accounted for the 46.637% of the total variance. The factor loads of the scale ranged from 0.41 to 0.70 and item total correlations from .474 to .668. Test–retest correlation of the scale was .771 and the correlation coefficient between the two halves of the scale was .685. The Cronbach's Alpha coefficient of the scale was .75 and the same coefficient has been calculated for this study, the resulting figure is 0.78.

Inquiry skills scale

The Inquiry Skills Scale was developed by Aldan Karademir and Saracaloğlu (2013) and it is a five-point Likert scale consisting of fourteen items and three sub-dimensions. The sub-dimensions are *information acquisition*, *information control* and *self-confidence*. All the items involved in the scale are positively worded. The KMO value of the scale was 0.854 and Barlett's test value was 1,513,582 ($p < .01$). The three factors in the scale accounted for the 51.98% of the total variance. The Cronbach's Alpha coefficients of the scale were .70 for information acquisition, .67 for information control, .90 for self-confidence and .79 for the overall scale. The Cronbach's Alpha coefficients calculated for each subdimension of this study were .70 for information acquisition, .67 for information control, .90 for self-confidence and finally .79 for the overall scale, respectively.

2.1.4 | Analysis of data

In order to examine the predictory power of inquiry skills on lateral thinking through path analysis, analyses were conducted for missing and extreme values. When the data set was investigated, it was observed that there were no missing values. Data belonging to 24 participants were extreme values. However, when the demographic data belonging to these participants were checked, it was seen that these individuals belonged to the sampling intended to draw from the population. So, they were included in the study. Also, univariate and multivariate analyses were carried out. Descriptive analyses (skewness, kurtosis), histograms, density and perspective graphics, Shapiro–Wilk and Mardia tests belonging to the variables were examined to test univariate and multivariate normality. The variables in the study consist of the mean values obtained from the lateral thinking scale and sub-dimensions of inquiry skills scale. Mean values of lateral thinking scale were named LaTh, and the sub-dimensions of inquiry skills scale were named as follows; information acquisition (InAc), information control (InCo), and self-confidence (SeCo). Descriptive statistics regarding these variables are given in Table 2.

In Table 2, skewness and kurtosis values provide information on the univariate normality of the data. These values must be either zero (0) or within acceptable intervals to be regarded normally distributed. There are varying values for acceptable intervals of skewness and kurtosis in the related literature like ±1, ±1.5, ±2. (Bulmer, 1979; Gravetter & Wallnau, 2014; Tabachnick & Fidell, 2013). When the proposed values on skewness and kurtosis are considered, it is not possible to make certain decisions on univariate normality and predict the multivariate

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TABLE 2 Descriptive statistics on the variables.

Variables	$\bar{X}$	SD	Skewness	Kurtosis
LaTh	3.775	0.534	−0.306	0.872
InAc	4.169	0.494	−0.821	2.063
InCo	3.646	0.707	1.359	17.630
SeCo	3.548	0.978	−0.307	−0.535

Note: $n = 975$.

TABLE 3 Shapiro–Wilk and Mardia tests for normality.

Test/Variable	Statistic	*p*
Shapiro–Wilk/LaTh	0.985	<.001
Shapiro–Wilk/InAc	0.953	<.001
Shapiro–Wilk/InCo	0.916	<.001
Shapiro–Wilk/SeCo	0.957	<.001
Mardia-Skewness	1397.129	<.001
Mardia-Kurtosis	100.427	<.001

normality in the study. Therefore, Shapiro–Wilk test for univariate normality and Mardia test for multivariate normality were conducted. The results are given in Table 3.

As the values belonging to both univariate and multivariate normality tests in Table 3 demonstrate, the data of the study are not normally distributed. However, the graphics of the variables were checked due to the fact that test results may be affected by the sample size.

Distribution curves are also added onto histogram graphics in Figure 1. As the distribution curves demonstrate, univariate normality has not been satisfied across all variables. To test multivariate normality, perspective graphics belonging to the binary combinations of variables are formed and given in Figure 2.

When perspective graphics are examined, it is observed that the joint distributions of the variables have exhibited skewed distributions away from the centres of the planes. As a result, it was found that multivariate normality has not been satisfied, either.

The results given above have a significant role in determining the method for the path analysis model. The statistical analysis was carried out using WLSMV (weighted least square mean and variance adjusted) estimation method as the data set were not normally distributed (Kline, 2011). Before the path analysis, reliability analysis was carried out. To calculate Cronbach Alpha reliability coefficient for sub-dimensions of inquiry skills scale and unidimensional lateral thinking scale, stratified Alpha was calculated as the Cronbach Alpha in inquiry skills scale required unidimensionality assumption (Crocker & Algina, 1986). In addition, multidimensional inquiry skills scale was analysed using divergent validity, HTMT (heterotrait-monotrait ratio of correlations). Fornell-Larcker criterion, frequently used to test divergent validity, was not preferred in the analysis as it has lower specificity and sensitivity ratios compared to HTMT (Henseler et al., 2015). The calculated values are given in Table 4.

As the values in Table 4 demonstrate that both lateral thinking scale and inquiry skills scale along with its subdimensions have medium and high levels of reliability (Guilford, 1946). When the HTMT, a measure for the similarity between latent variables, was examined for the subdimensions of multidimensional inquiry skills scale, it was decided that divergent validity was satisfied as all the values were lower than 1 (Ab Hamid et al., 2017). Regarding divergent validity, Kline (2011) claims that cut-off values of the coefficients should be lower than 0.85. Even so, divergent validity is satisfied.

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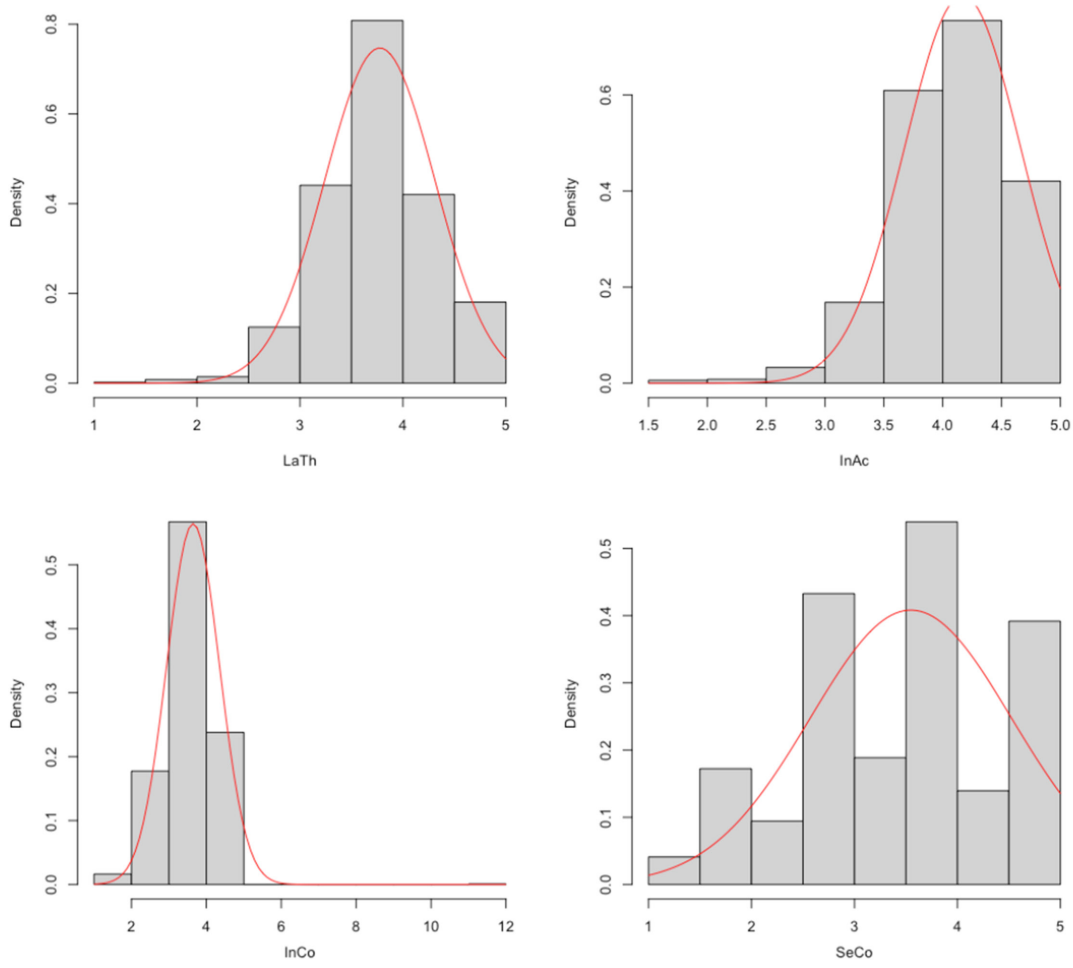


FIGURE 1 Histogram graphics of the variables.

All the analyses were carried out using R Studio. In the analyses, haven (Wickham & Miller, 2021), MVN (Korkmaz et al., 2014), sirt (Robitzsch, 2021), lavaan (Rosseel, 2012), semPlot (Epskamp, 2022) and semTools (Jorgensen et al., 2021) packages were used.

3 | FINDINGS AND INTERPRETATIONS

The findings obtained in the study are given below in line with the three questions listed under the title *Purpose of the Study*. Thus, the mean and standard deviation values concerning the prospective teachers' inquiry skills and lateral thinking dispositions are given in Table 5 as the answer of the first question, what is the level of prospective teachers' inquiry skills and lateral thinking dispositions?

As the data in Table 5 demonstrate that the mean values of information acquisition ($\overline{X}=4.17$), information control ($\overline{X}=3.65$), self-confidence ($\overline{X}=3.55$) and overall scale ($\overline{X}=3.85$) are high as the score intervals for the categories in a five-point Likert scale are; strongly disagree: 1.00–1.79, disagree: 1.80–2.59, neutral: 2.60–3.39, agree: 3.40–4.19 and strongly agree: 4.20–5.00. Thus, students usually *agree* with the items under all categories. It can be stated that students use their inquiry skills most while revising their decisions, selecting the necessary

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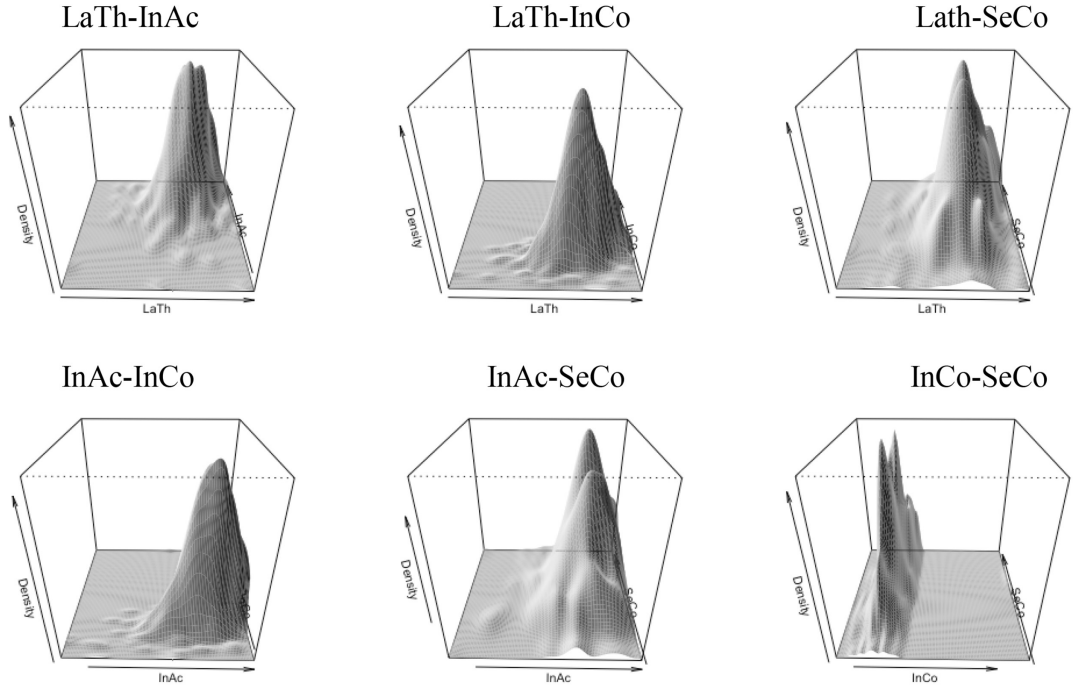


FIGURE 2 Perspective graphics belonging to the binary combinations of variables.

TABLE 4 Reliability coefficients belonging to data collection instruments.

Variable	Cronbach alpha	Stratified alpha	HTMT		
			InAc	InCo	SeCo
LaTh	.781				
InqS		.842			
InAc	.700		–		
InCo	.666		.628	–	
SeCo	.901		.469	.565	–

Abbreviation: InqS, inquiry skills scale.

information, listening to others' replies, using previous information for a different problem and correcting their mistakes under the subdimension *information acquisition*. The category with the second highest mean score is *information control*. Under this category, students generally agree with the idea that they gather materials, experiences and objects to solve a problem, they test the accuracy of what they read, they look for alternative solutions even if they find a solution, they compare the information obtained through individual experiences with the information they get from more than one sources and they try various solutions to answer a question. Having the lowest mean score among the three subdimensions, *self-confidence* involves speaking out one's thoughts about what is taught in the class, asking about unclear points in a discussion and asking the teacher about the topic which is hard to comprehend. Table 5 also shows that lateral thinking dispositions of teachers are high ($\overline{X} = 3.78$). The skills involved in lateral thinking range from producing alternative solutions for problems to looking at events from different perspectives and dealing with seemingly improbable approaches while thinking. Thus, the fact that prospective teachers' inquiry skills and lateral thinking dispositions are high could be deemed positive as they are

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TABLE 5 Mean and standard deviation values regarding prospective teachers' inquiry skills and lateral thinking dispositions.

Scales	Subdimensions	$\overline{X}$	SD
Inquiry skills	Information acquisition	4.17	.49
	Information control	3.65	.71
	Self confidence	3.55	.98
	Overall scale	3.85	.50
Lateral thinking disposition	Overall scale (unidimensional)	3.78	.53

expected to acquire these skills at university and enable their prospective students to employ inquiry skills and lateral thinking while handling an issue or solving a problem.

One of the main aims of this study is to identify the effect of the departments in which students were enrolled on inquiry skills and lateral thinking dispositions of students. So, one-way ANOVA test was carried out on the mean values of students studying in different departments. Table 6 shows whether students' mean scores vary significantly with respect to the department they attend.

As shown in Table 6, inquiry skills of prospective teachers differ significantly in terms of the departments they attend in the overall scale ($F=3.004$; $p<.05$) as well as in the subdimensions "information control" ($F=3.635$; $p<.05$), and self-confidence ($F=2.929$; $p<.05$), whereas no significant difference is observed in the subdimension "information acquisition" ($F=1.169$; $p>.05$). It is noteworthy that in all the categories where a significant difference is observed, the lowest mean score always belongs to the students in the department of mathematics education. This means that the students in mathematics education don't use their inquiry skills as much as those in other departments. Inquiry skills of students at the department of science education ($\overline{X}=3.83$) and mathematics education ($\overline{X}=3.49$) differ significantly in the subdimension "information control". As for the subdimension "self-confidence", significant differences were observed in the inquiry skills between the students in the departments of mathematics education ($\overline{X}=3.34$) and those in science education ($\overline{X}=3.75$). As for the lateral thinking dispositions of prospective teachers, significant differences were observed ($F=2{,}408$; $p<.05$) between the mean scores of students in mathematics education ($\overline{X}=3.69$) and those in Turkish language ($\overline{X}=3.82$) and science education ($\overline{X}=3.84$). Again, the students in mathematics education lacked lateral thinking skills when compared to other departments as is the case with inquiry skills, which is the most remarkable finding in this part.

An analysis of variance test (ANOVA) was conducted to see whether the mean scores of students differ with respect to class level. Table 7 shows the findings obtained through ANOVA test.

When the above data are analysed, it is observed that inquiry skills scores of students differ significantly in terms of class levels only in the subdimension "information acquisition". No other significant difference was identified between groups in the analysis. First year students ($\overline{X}=4.25$) use inquiry skills more than the fourth year students ($\overline{X}=4.09$) in the subdimension "information acquisition" ($F=3.599$; $p<.05$). As for the lateral thinking dispositions of students, the mean scores of students do not differ ($F=1.643$; $p>.05$) in terms of class levels. Thus, it could be claimed that students' lateral thinking disposition scores are close to each other and the years spent in higher education do not influence the lateral thinking dispositions of students.

To determine the relation between inquiry skills and lateral thinking dispositions of students, a structural model was established, and a series of fit indices were used to test the adequacy of the model fit. Some researchers suggested noble and acceptable fit indices to be used for model fit adequacy (DiStefano et al., 2018; Hu & Bentler, 1999; Kline, 2015; Tabachnick & Fidell, 2001). The suggested fit indices are given in Table 8. In addition to frequently used fit indices such as CFI, RMSEA, SRMR values, WRMR value was also included as WLSMV was used as the estimation method in the study.

It could be stated that the model used in the study is covarince-based SEM (CB-SEM) as a path analysis model was formed to test the hypothesis that inquiry skills affect lateral thinking. The model is given in Figure 3. Before

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TABLE 6 ANOVA test results displaying the difference of mean scores in terms of the departments students attend.

Scales		Departments	*N*	$\overline{X}$	SD	df	F	p	Scheffe/LSD
	Subdimensions								
Inquiry skills		1. Turkish	267	4.22	.51		1.169	.323	
	Information acquisition	2. Science	85	4.11	.41	4			
		3. Mathematics	162	4.14	.52	970			
		4. Elementary Ed.	302	4.16	.44	974			
		5. Social Sciences	159	4.16	.57				
	Information control	1. Turkish	267	3.68	.65		3.635	.006*	(Scheffe)
		2. Science	85	3.83	.66	4			2–3
		3. Mathematics	162	3.49	.62	970			
		4. Elementary Ed.	302	3.65	.78	974			
		5. Social Sciences	159	3.65	.74				
	Self-confidence	1. Turkish	267	3.55	.98		2.929	.020*	(Scheffe)
		2. Science	85	3.75	.88	4			2–3
		3. Mathematics	162	3.34	.97	970			
		4. Elementary Ed.	302	3.56	.98	974			
		5. Social Sciences	159	3.60	1.00				
	Overall scale	1. Turkish	267	3.88	.50		3.004	.018*	(LSD)
		2. Science	85	3.93	.45	4			1–3
		3. Mathematics	162	3.74	.48	970			2–3
		4. Elementary Ed.	302	3.85	.48	974			3–4
		5. Social Sciences	159	3.86	.56				3–5
Lateral thinking disposition		1. Turkish	267	3.82	.53		2.408	.048*	(LSD)
		2. Science	85	3.84	.46	4			1–3
		3. Mathematics	162	3.69	.57	691			2–3
		4. Elementary Ed.	302	3.74	.52	695			
		5. Social Sciences	159	3.81	.54				

*$p < .05$.

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TABLE 7 ANOVA test showing the difference of prospective teachers' mean scores in inquiry skills and lateral thinking dispositions in terms of class level.

		Classes	*N*	$\overline{X}$	SD	Df	F	p	Scheffe
Inquiry skill	*Subdimensions*								
	Information acquisition	1st year	226	4.25	.48		3.599	.013*	1–4
		2nd year	391	4.16	.50	3			
		3rd year	194	4.17	.51	971			
		4th year	164	4.09	.48	974			
	Information control	1st year	226	3.63	.86		0.117	.950	
		2nd year	391	3.64	.66	3			
		3rd year	194	3.67	.69	971			
		4th year	164	3.64	.61	974			
	Self confidence	1st year	226	3.47	.96		0.704	.550	
		2nd year	391	3.58	.98	3			
		3rd year	194	3.54	1.01	971			
		4th year	164	3.59	.94	974			
	Overall scale	1st year	226	3.86	.52		0.260	.854	
		2nd year	391	3.85	.49	3			
		3rd year	194	3.86	.52	971			
		4th year	164	3.82	.47	974			
Lateral thinking disposition		1st year	226	3.75	.57		1.643	.178	
		2nd year	391	3.74	.52	3			
		3rd year	194	3.82	.52	971			
		4th year	164	3.83	.54	974			

*$p < .05$.

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studying the fit indices belonging to the model in Figure 3, it can be stated that model-data fit is realised when the *p* value ($p = .314$) that tests statistical significance of observed and determined covariance matrix is considered. Moreover, when the fit indices (CFI = 0.999, RMSEA = 0.013, SRMR = 0.018, WRMR = 0.481) are examined and compared with the criterion values in Table 8, it can be stated that a perfect model-data fit is achieved. Finally, it was found out that there was not a significant proposal when the modification indices regarding the model were analysed and the model displayed in Figure 3 was agreed upon.

As Figure 3 demonstrates, inquiry skills which affect lateral thinking—the observed variable in the model- are added into the model as the latent variable. The observed structures which account for inquiry skills are information acquisition, information control and self-confidence. The parameters and values concerning the direct effects are given in Table 9.

As the values in Table 9 indicate, all the path coefficients except for one (InqS → InAc) are significant. However, significance of the coefficient isn't tested due to the fact that this path coefficient is fixed to 1 so as to determine the scale of InqS latent variable. When the standardised values of the coefficients are examined, it is observed that only InqS → SeCo path coefficient is lower than .50 value, proposed by Byrne (2013). But, SeCo observed variable hasn't been removed from the model for three reasons. This standardised path coefficient (.467) is quite close to cut-off value, it is indicated in the related literature that the observed variable, self-confidence (SeCo)

TABLE 8 Criterion values for good and acceptable fit.

Indices	Noble fit	Acceptable fit
p value	$p > .05$	$p > .05$
CFI	$.95 \leq CFI \leq 1$	$0.90 \leq CFI < 0.95$
RMSEA	$.00 \leq RMSA \leq .05$	$.05 < RMSA \leq .08$
SRMR	$.00 \leq SRMR \leq .05$	$.05 < SRMR \leq .10$
WRMR	$WRMR < 0.9$	$WRMR < 1$

Abbreviations: CFI, comparative fix index; RMSEA, root mean error of approximation; SRMR, standardised root mean square residual; WRMR, weighted root mean square residual.

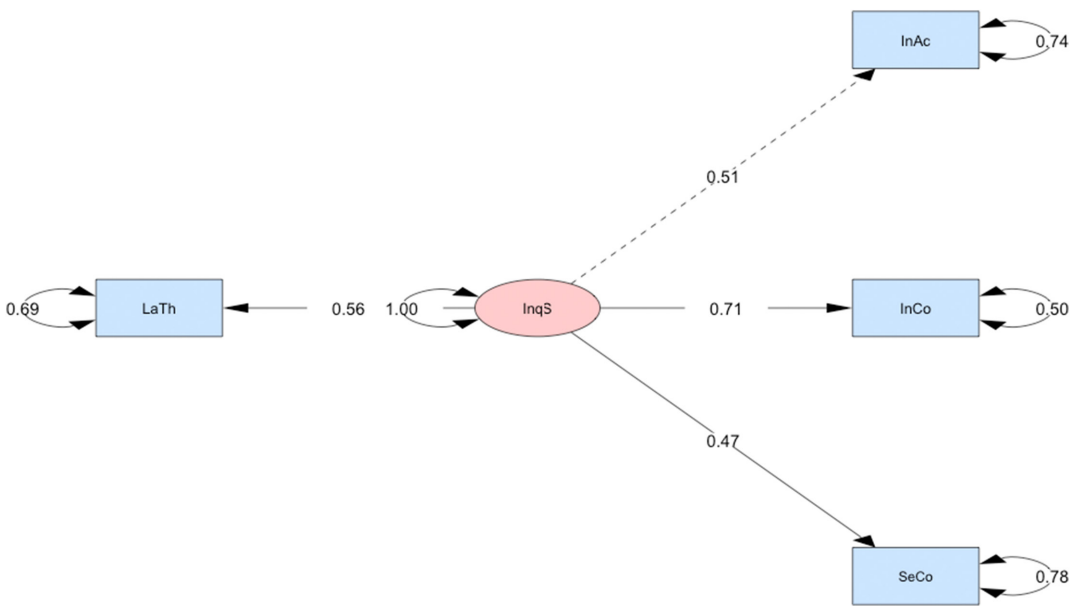


FIGURE 3 Path analysis model.

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TABLE 9 Results of path analysis model parameters.

Parameter	Unstandardized	SE	Standardised
Direct effects			
InqS → LaTH	0.298*	0.148	0.558
InqS → InAc	0.251	–	0.507
InqS → InCo	0.501*	0.189	0.709
InqS → SeCo	0.456*	0.2217	0.467
Variances			
InAc	0.182	0.013	0.743
InCo	0.248	0.078	0.497
SeCo	0.747	0.040	0.782
LaTh	0.197	0.015	0.689
InqS	0.063	0.011	1.000

*$p < .01$.

accounts for inquiry skills (Aldan Karademir & Saracaloğlu, 2013) and the model turns out to be just identification (SD = 0) when the observed variable, SeCo is removed from the model.

In conclusion, the effect of inquiry skills on lateral thinking is 0.558. This value indicates that one-point standard deviation increase in inquiry skills raises the standard deviation of the predicted lateral thinking at a rate of 0.589. The standardised variance ratio belonging to lateral thinking is calculated 0.689. In other words, almost 69% of the total variance for lateral thinking isn't explained. To put it another way, inquiry skills account for 31% (558^2) of lateral thinking.

4 | RESULTS, DISCUSSION AND RECOMMENDATIONS

It was stressed earlier in this study that lateral thinking and inquiry skills are essential skills to be taught to students to prepare them for the challenges of 21st century. In an educational context, prospective teachers are the right group to focus on when teaching these two skills are mentioned. They are still students who need to improve their skills and teachers of the future who are expected to teach these skills to their students. Therefore, having an idea about their lateral thinking and inquiry skills could shed light on the acquisition of these skills in the future.

This study aimed at measuring the inquiry skills and lateral thinking dispositions of higher education students who were enrolled in various departments at three faculties of education and examining the relations between these two variables. Then, it was revealed whether students' mean scores differ with respect to variables such as department and class level. As the findings in this study suggest, prospective teachers' lateral thinking skills are relatively high with an approximate percentage of 70%. This finding can be regarded positive as lateral thinking is associated with creative thinking, which is one of the 21st century skills. As Karagöz (2019) indicates, lateral thinking implementations, which enable prospective teachers to realise their real potentials and learn at their own pace should be incorporated into the instructional processes. Also, the fact that prospective teachers can think laterally, breaking the habitual thinking patterns employed for a long time, can raise hopes for future generations as the teachers who are expected to teach thinking including lateral thinking have already had high levels of lateral thinking. This finding is supported by those obtained in the recent studies carried out by Yıldız and Yılmaz (2020), Yazgan (2021) and Elaldı (2021). They also found out that higher education students' lateral thinking levels are high. On the other hand, Lawrence and Xavier (2013) observed that prospective teachers' level of lateral thinking was moderate. As opposed to Lawrence and Xavier's study, the three studies which have similar findings with

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those obtained in this study were carried out in similar contexts and on similar participants, which could explain the similarity of findings.

It was also found out that prospective teachers' level of inquiry skills was high. Having high levels of inquiry skills means that the individuals don't suffice with what they have, think critically and logically and extend their knowledge through asking questions and connecting new and existing knowledge. These skills are extremely valuable for future teachers as the individuals living in the 21st century need to learn thinking skills including critical thinking and inquiry skills (Partnership for 21st Century Skills, 2009). Katwijk et al. (2021) point out in their study that all participants, students and educators believe inquiry is valuable and think that learning outcomes of inquiry is enriching. Cakmak et al. (2016) found out in their study that prospective teachers' levels of inquiry skills in the department of social studies were high, which corresponds to the finding of this study. In addition, Balbağ and Aynur (2020) studied the inquiry skills of prospective science teachers and they stated that prospective science teachers' levels of inquiry skills were above average. Both of these studies used the same scale used in this study and in similar contexts and the results were similar too.

It was discovered in this study that lateral thinking dispositions of prospective teachers differed significantly in terms of the departments they attended, which resulted from the relatively low mean score of the students in mathematics education department. Similarly, in the study carried out by Lawrence and Xavier (2013) in India, it was found out that department was a significant determinant on lateral thinking levels of prospective teachers. The findings of another study carried out by Semerci (2017) also confirm that of this study. Semerci indicated that prospective teachers' lateral thinking dispositions differed significantly with regards to departments in which students were enrolled. When the fact that skills and knowledge in different areas are required to be accepted to different departments at a higher education institution is taken into consideration, it would be reasonable to expect differing levels of lateral thinking between departments, especially between physical and social sciences. As for inquiry skills, significant differences were observed between the mean scores of students attending different departments. The most remarkable point to stress here is that students at mathematics education department always got the lowest scores except for the subdimension *information acquisition* and significant differences were observed between the mean scores of students at mathematics education and those of students at other departments. Şimşek and Soytok (2016) compared pre-service teachers' (music and science education departments only) levels of inquiry skills and they did not find significant differences between the students at the two departments although the mean scores of the students at music education department were higher. There were no participants from the department of music education in our study and all the differences resulted from the low scores of the students at mathematics education department. In other words, there may not be significant differences depending on the departments if the students at mathematics education department weren't included in the study, which is also valid for lateral thinking dispositions of students.

Another finding obtained in this study was that lateral thinking disposition scores of prospective teachers did not differ in terms of class levels. Similarly, Karagöz (2019) and discovered that seniority was not a determining factor in prospective teachers' lateral thinking dispositions. On the other hand, Yıldız and Yılmaz (2020) found out in their study that seniority meaningfully affected the lateral thinking dispositions of prospective teachers. When universities are regarded as the institutions that promote thinking skills including lateral thinking, significant improvement can be expected depending on the time students spent at higher education institutions, which could account for significant differences between students' scores in lateral thinking. However, unless universities develop students' thinking skills, the time spent at those institutions will possibly not affect lateral thinking dispositions of students, which could explain the findings of this study. As for inquiry skills of students, again, no significant differences were observed between students' mean scores except for the subdimension, *information acquisition*. First and fourth year students differed significantly regarding their inquiry skills under this subdimension. However, Cakmak et al. (2016) revealed in their study that prospective social sciences teachers' inquiry skills changed significantly with respect to seniority. This difference between the two studies may result from the fact that this study is more comprehensive, including five departments. As the overall mean scores of students are

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usually high, the fact that students' inquiry skills don't differ significantly could be interpreted in such a way that the university doesn't develop students' inquiry skills considerably or the students were very well equipped with inquiry skills when they were at high school.

Finally, the main finding of the study is that inquiry skills predict lateral thinking dispositions at a percentage of 31%. In other words, lateral thinking dispositions of pre-service teachers depend on their inquiry skills at a percentage of 31%. In the literature, no research was found studying the predictive relations between inquiry skills and lateral thinking, which makes this study necessary and valuable. Researchers rarely studied the relation between lateral thinking and other variables and Yazgan (2021) is one of them. Yazgan studied the predictive power of problem-solving on pre-service teachers' lateral thinking. It was revealed in the study that pre-service teachers' lateral thinking levels were high, whereas their problem-solving skills were moderate and problem-solving skills of the participants significantly predicted their lateral thinking. However, there are several studies on the relation between inquiry skills and types of thinking, especially critical thinking. Ghaemi and Ghazi (2017) indicated in their experimental study that inquiry-based activities significantly influenced critical thinking skills of participants. Similarly, Duran and Dökme (2016) indicated in their study that inquiry-based learning affects critical thinking skills. Another study examining the relation between inquiry and critical thinking skills was carried out by Qing et al. (2010). They stated that inquiry-based chemical experiment activity considerably developed pre-service teachers' critical thinking skills. In addition, Seranica et al. (2018) discovered in their study that guided inquiry model had a significant effect on critical thinking skills. The relation between inquiry skills and critical thinking shouldn't be surprising as inquiry requires thinking critically. Al-afifi and Ambusaidi (2014) stated that there was a positive correlation between students' inquiry skills and logical thinking skills. Moreover, Damopolii et al. (2021) concluded in their study that inquiry-based science learning activities positively affected students' thinking levels. The final study examining the relation of inquiry skills with the types of thinking was conducted by Madhuria, Kantamreddia and Prakash Gotetib (Madhuri et al., 2012). They pointed out that inquiry-based learning developed higher order thinking skills. In a nutshell, all the studies cited in this paragraph demonstrate that inquiry skills have positive effects on thinking skills, which corresponds to the findings obtained in this study.

5 | RECOMMENDATIONS

In the light of all the findings in the research, some recommendations were given. As students' inquiry skills predict their lateral thinking dispositions at a percentage of 35%, inquiry skills of students should be developed so as to promote their lateral thinking, which could also improve creative thinking and problem-solving skills of students. It is good news that higher education students' levels of inquiry skills and lateral thinking skills are high. However, a group of students attending the department of mathematics education scored usually low during the implementation. The reasons behind the low scores of students at mathematics education department regarding inquiry skills should be identified and some classes should be incorporated into the curricula to foster their inquiry skills. Also, higher education institutions should encourage their students to think critically, creatively and laterally in addition to using inquiry whenever they acquire information. In addition to encouragement, universities should deliberately promote their students' thinking skills through elective or compulsory classes or extracurricular activities.

6 | IMPLICATIONS FOR FUTURE RESEARCH

This study was a descriptive one. An experimental study on the effect of inquiry based learning activities on lateral thinking disposition of students should be carried out to observe the effect of practice rather than opinions of students. Again, rather than prospective teachers, the relations between teachers' inquiry skills and lateral thinking dispositions can be studied. As there are few studies on the relation between lateral thinking skills and

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other types of thinking, the relations between lateral thinking skills and various higher order thinking skills can be studied. Finally, the distribution of the participants was not balanced in this study with respect to gender as only one-fourth of the participants consisted of female students. It would be better to conduct a study with a more balanced distribution in terms of the gender of the participants.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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REFERENCES

Ab Hamid, M. R., Sami, W., & Sidek, M. M. (2017, September). Discriminant validity assessment: Use of Fornell & Larcker criterion versus HTMT criterion. *Journal of Physics: Conference Series*, *890*(1), 012163 IOP Publishing.

Abalı Öztürk, Y., Bilgen, Z., & Bilgen, S. (2017). Relationship between inquiry and self-directed learning skills: An investigation on the elementary education prospective teachers. *Sinop University Journal of Social Science*, *1*(2), 179–214. https://doi.org/10.7827/TurkishStudies.12096

Al-afifi, M., & Ambusaidi, A. K. (2014). The relationship between level of inquiry skills and logical thinking abilities among tenth grade omani students. *An-Najah University Journal for Research - Humanities*, *28*, 2521–2556.

Aldan Karademir, Ç., & Saracaloğlu, A. S. (2013). The development of inquiry skills scale: Reliability and validity study. *Asian Journal of Instruction*, *1*(2), 56–65.

Aldan Karademir, Ç., & Saracaloğlu, A. S. (2017). The effect of pre-service teachers' inquiry and critical thinking skills on teacher self efficacy. *Journal of Turkish Studies*, *12*(33), 261–290. https://doi.org/10.7827/TurkishStudies.12708

Alkhayari, H. (2021). *Fostering 21st century learning skills in English as a second language classrooms* (Master's thesis). Department of Integrated Studies in Education, McGill University.

Arslan, H., Ünver, Ş., & Yılmaz, M. (2022). Investigation of empathic tendency and lateral thinking tendency skills of faculty of sport sciences students. *Mediterranean Journal of Sport Science*, *5*(3), 499–509.

Aslan, S. (2017). Learning by teaching: Can it be utilized to develop ınquiry skills? *Journal of Education and Training Studies*, *5*(12), 190–198.

Aydemir, A. (2021). Social innovation and lateral thinking tendencies of preservice social studies teachers. *International Journal of Progressive Education*, *17*(3), 250–262.

Bailey, L. A. (2018). *The impact of inquiry-based learning on academic achievement in eighth-grade social studies* [Unpublished doctoral dissertation]. University of South Carolina.

Balbağ, M. Z., & Aynur, D. (2020). Investigation of science teacher candidates' ınquiry skills in terms of some variables. *International Journal of Social Science Research*, *9*(1), 48–62. https://www.researchgate.net/publication/342534247

Baron, J. (1993). Why teach thinking? An essay. *Applied Psychology: An International Review*, *42*(3), 191–237.

Baron, J. (2008). *Thinking and deciding* (4th ed.). Cambridge University Press.

Bedir, T., & Duman, B. (2017). Examining the inquiry skills of pre-service teachers. *Electronic Turkish Studies*, *12*(18), 105–120. https://doi.org/10.7827/TurkishStudies.12096

Bell, R. L., Smetana, L., & Binns, I. (2005). Simplifying inquiry instruction. *The Science Teacher*, *72*(7), 30–33. https://www.researchgate.net/publication/228665515

Brewerton, K. (2010). *Leading the development of thinking for understanding within inquiry* (Unpublished MA thesis). Unitec Institute of Technology. https://hdl.handle.net/10652/1607

Bulmer, M. G. (1979). *Principles of statistics*. Dover.

Burden, R. (1998). How can we best help children to become effective thinkers and learners? In R. Burden & M. Williams (Eds.), *Thinking through the curriculum*. Routledge.

Burgh, G. (2014). Creative and lateral thinking: Edward de Bono. In D. C. Phillips (Ed.), *Encyclopedia of educational theory and philosophy* (pp. 187–189). Sage Publications. https://doi.org/10.4135/9781483346229.n86

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Byrne, B. M. (2013). *Structural equation modeling with Mplus: Basic concepts, applications and programming*. Routledge.
Cakmak, Z., Taşkıran, C., & Bulut, B. (2016). A study on the inquiry skills of social studies teacher candidates. *International Journal of Learning and Teaching*, *2*(1), 58–62. https://doi.org/10.18178/IJLT.2.1.58-62
Chang, Z., & Zhou, M. (2022). The influence of different music styles on Chinese students' lateral thinking skills. *Thinking Skills and Creativity*, *43*, 1–9. https://doi.org/10.1016/j.tsc.2021.100990
Costa, A. (1991). The inquiry strategy. In A. Costa (Ed.), *Developing minds: A source book for teaching thinking*. Association for Supervision and Curriculum Development. https://files.eric.ed.gov/fulltext/ED332166.pdf
Creswell, J. W. (2003). *Research design: Qualitative, quantitative. And mixed methods approaches* (2nd ed.). SAGE Publications.
Crocker, L., & Algina, J. (1986). Introduction to classical and modern test theory. Holt, Rinehart and Winston.
Csapo, B. (2001). Cognitive aspects of democratic thinking. In R. Soder, J. Goodlad, & T. J. McMannon (Eds.), *Developing democratic character in the young*. Jossey-Bass. https://www.academia.edu/9206802
Damopolii, I., Nusantari, E., Kristianto, D. D., Nunaki, J. H., & Kandowangko, N. Y. (2021). The effect of inquiry-based science learning on students' level of thinking. *Journal of Physics: Conference Series. Mathematics and Science Education International Seminar (MASEIS)*, *1731*, 012089. https://doi.org/10.1088/1742-6596/1731/1/012089
Davies, D., Collier, C., & Howe, A. (2012). Assessing scientific and technological enquiry skills at age 11 using the e-scape system. *International Journal of Technology and Design Education*, *22*, 247–263. https://www.learntechlib.org/p/64590/
De Bono, D. (1990). *Lateral thinking: A textbook of creativity*. Penguin Books.
De Bono, E. (1971). *The use of lateral thinking*. Penguin Books.
De Bono, E. (1977). *Lateral thinking: A textbook of creativity*. Penguin Books.
De Bono, E. (1979). *The use of lateral thinking: Break the stranglehold of logical thinking*. GB: Pelican Books.
De Bono, E. (2014). *Lateral thinking: An introduction*. Vermilion.
DiStefano, C., Liu, J., Jiang, N., & Shi, D. (2018). Examination of the weighted root mean square residual: Evidence for trustworthiness? *Structural Equation Modeling: A Multidisciplinary Journal*, *25*(3), 453–466.
Dolapçıoğlu, S. (2020). Promoting inquiry skills for the improvement of self-efficacy domains in special education pre-service teacher training. *Journal of Teacher Education and Educators*, *9*(2), 201–220. https://files.eric.ed.gov/fulltext/EJ1272766.pdf
Dow, P. (2000). Why inquiry? A historical and philosophical commentary. Foundations 'A monograph for professionals in science, mathematics, and technology education'. *National Science Foundation*, *2*, 5–9.
Duran, M., & Dökme, I. (2016). The effect of the inquiry-based learning approach on student's critical-thinking skills. *Eurasia Journal of Mathematics, Science & Technology Education*, *12*(12), 2887–2908. https://doi.org/10.12973/eurasia.2016.02311a
Elaldı, Ş. (2021). Effect of lateral thinking dispositions on groupwork. *Journal of Technical Education and Training*, *13*(2), 74–94. https://publisher.uthm.edu.my/ojs/index.php/JTET/article/view/6539
El Soufi, N., & See, B. H. (2019). Does explicit teaching of critical thinking improve critical thinking skills of English language learners in higher education? A critical review of causal evidence. *Studies in Educational Evaluation*, *60*, 140–162.
Epskamp, S. (2022). semPlot: Path diagrams and visual analysis of various SEM packages' output. R package version 1.1.5. https://CRAN.R-project.org/package=semPlot
Fraenkel, J. R., Wallen, N. E., & Hyun, H. H. (2012). *How to design and evaluate research in education* (8th ed.). McGraw Hill. ISBN-13: 978-0-07-809785-0.
Ghaemi, F., & Ghazi, J. (2017). The impact of inquiry-based learning approach on critical thinking skill of EFL students. *EFL Journal*, *2*(2), 89–102. https://doi.org/10.21462/eflj.v2i2.38
Ghasemy, M., Elwood, J. A., & Scott, G. (2023). A comparative study on turnaround leadership in higher education and the successful implementation of the UN's sustainable development goals. *International Journal of Sustainability in Higher Education*, *24*(3), 602–636.
Gormally, C., Brickman, P., Hallar, B., & Armstrong, N. (2009). Effects of inquiry-based learning on students' science literacy skills and confidence. *International Journal for the Scholarship of Teaching and Learning*, *3*(2), Article 16. https://doi.org/10.20429/ijsotl.2009.030216
Gravetter, F., & Wallnau, L. (2014). *Essentials of statistics for the behavioral sciences*. Wadsworth.
Guilford, J. P. (1946). New standards for test evaluation. *Educational and Psychological Measurement*, *6*(4), 427–438.
Güneş, F. (2012). Improving the thinking skills of students. *Journal of Turkology Research*, *32*, 127–146. https://dergipark.org.tr/tr/pub/tubar/issue/16973/177364
Halpern, D. F. (1997). *Critical thinking across the curriculum: A brief edition of thought and knowledge*. Erlbaum Associates, Inc.
Harlen, W. (2013). Inquiry-based learning in science and mathematics. *Review of Science, Mathematics and ICT Education*, *7*(2), 9–33. https://doi.org/10.26220/REV.2042

14653435, 2024, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/ejed.12673, Wiley Online Library on [27/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Harrison, C. (2014). Assessment of inquiry skills in the sails project. *Science Education International*, *25*(1), 112–122. http://www.icaseonline.net/sei/march2014/p12.pdf

Hayat, M. S. & Rustaman, N. Y. (2017). How is the inquiry skills of biology preservice teachers in biotechnology lecture? International conference on mathematics and science education (ICMScE). IOP Conference Series. *Journal of Physics: Conference Series 895*, 012135. https://doi.org/10.1088/1742-6596/895/1/012135

Henseler, J., Ringle, C. M., & Sarstedt, M. (2015). A new criterion for assessing discriminant validity in variance-based structural equation modeling. *Journal of the Academy of Marketing Science*, *43*(1), 115–135.

Honzel, B. (2012). *The effects of an inquiry-based data-to-concept curriculum* (Unpublished MSc thesis). Montana State Universisty.

Hu, L., & Bentler, P. M. (1999). Cutoff criteria for fit indexes in covariance structure analysis: Conventional criteria versus new alternatives. *Structural Equation Modeling: A Multidisciplinary Journal*, *6*(1), 1–55. https://doi.org/10.1080/10705519909540118

Joreskog, K., & Sörbom, D. (2001). *Lisrel 8: User's reference guide*. Scientific Software International Inc. ISBN-13: 978-0894980404 ISBN-10: 0894980408.

Jorgensen, T. D., Pornprasertmanit, S., Schoemann, A. M., & Rosseel, Y. (2021). semTools: Useful tools for structural equation modeling. R package version 0.5-5. https://CRAN.R-project.org/package=semTools

Julita, D., & Herman, T. (2019). Improving mathematical lateral thinking ability of high school students through quantum learning based on creative problem solving. International Seminar on Applied Mathematics and Mathematics Education 2019 IOP Conference Series. *Journal of Physics: Conference of Series*, *1315*, 012061. IOP Publishing. https://doi.org/10.1088/1742-6596/1315/1/012061

Karagöz, B. (2019). An evaluation of pre-service turkish teachers' lateral thinking dispositions with regard to different variables. *European Journal of Education Studies*, *5*(9), 156–164. https://doi.org/10.5281/zenodo.2542955

Katwijk, L., Jansen, E., & Veen, K. (2021). Development of an inquiry stance? Perceptions of preservice teachers and teacher educators toward preservice teacher inquiry in Dutch primary teacher education. *Journal of Teacher Education*, *73*, 286–300. https://doi.org/10.1177/00224871211013750

Kline, R. B. (2011). *Principles and practice of structural equation modeling* (3rd ed.). The Guilford Press.

Kline, R. B. (2015). *Principles and practice of structural equation modeling*. Guilford Publications. ISBN 9781462523344.

Korkmaz, S., Goksuluk, D., & Zararsiz, G. (2014). MVN: An R package for assessing multivariate normality. *The R Journal*, *6*(2), 151–162.

Kumari, S., & Aggarwal, M. (2012). Intelligence and achievement as the correlates of lateral thinking of the student teachers. *International Indexed & Referred Research Journal*, *4*(41), 31–33. https://www.academia.edu/1253794/32_33

Kurnaz, A. (2013). *Eleştirel Düşünme Öğretimi Etkinlikleri*. Eğitim Yayınevi.

Lawrence, A. S. A., & Xavier, S. A. (2013). Lateral thinking of prospective teachers. *Journal of Educational Reflection*, *1*(1), 28–32. ISSN 2320-3145.

Lim, B. R. (2001). *Guidelines for designing inquiry-based learning on the web: Online professional development of educators*, PhD thesis. Indiana University. https://www.learntechlib.org/p/124113/

Lin, J. (2007). A study of students' ınquıry-based ı-search learnıng experıences. (Doctor of Philosophy [PHD]), University of Georgia; Winter 2007. https://getd.libs.uga.edu/pdfs/lin_jing_200712_phd.pdf

Lu, K., Yang, H. H., Shi, Y., & Wang, X. (2021). Examining the key influencing factors on college students' higher-order thinking skills in the smart classroom environment. *International Journal of Educational Technology in Higher Education*, *18*(1), 1–13.

MacLean, S., & Gray, K. (1998). Structural equation modeling in market research. www.Smalwaters.com/whitepapers/marketing, downloaded from 20.10.2021

Madhuri, G. V., Kantamreddia, V. S. S. N., & Prakash Gotetib, L. N. S. (2012). Promoting higher order thinking skills using inquiry-based learning. *European Journal of Engineering Education*, *37*, 117–123. https://doi.org/10.1080/03043797.2012.661701

Magnussen, L., Ishida, D., & Itano, J. (2000). The impact of the use of inquiry-based learning as a teaching methodology on the development of critical thinking. *Journal of Nursing Education*, *39*(8), 360–364. https://doi.org/10.3928/0148-4834-20001101-07

Martin-Hansen, L. (2002). Defining inquiry: Exploring the many types of inquiry in the science classroom. *Science Teacher*, *69*(2), 34–37. https://www.researchgate.net/publication/287494829

Mayer, R. (1983). *Thinking, problem solving, cognition*. Freeman. https://www.worldcat.org/title/thinking-problem-solving-cognition/oclc/23731853

McTighe, J., & Schollenberger, J. (1991). Why teach thinking? A statement of rationale. In A. L. Costa (Ed.), *Developing minds: A resource book for teaching thinking* (pp. 1–5). ASCD. https://files.eric.ed.gov/fulltext/ED332166.pdf

Moseley, D., Baumfield, V., Elliott, J., Gregson, M., Higgins, S., Miller, J., & Newton, D. P. (2005). *Frameworks for thinking: A handbook for teaching and learning*. Cambridge University Press. https://doi.org/10.2277/0521848318

14653435, 2024, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/ejed.12673, Wiley Online Library on [27/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Myers, K. B. (2019). *Changes in advanced educators' conceptualization of and concerns about global education as a result of participation in formal professional development* (Doctoral dissertation). Graduate Faculty of Texsas Tech University.

Nageswari, R., Ravikumar, B., & Stephen Jayamani, T. (2016). Building lateral thinking strategies to impart english speaking skills. *Pertanika J. Soc. Sci. & Hum.*, *24*(2), 687–700. http://www.pertanika.upm.edu.my/resources/files/Pertanika%20PAPERS/JSSH%20Vol.%2024%20(2)%20Jun.%202016/08%20JSSH%20Vol%2024%20(2)%20Jun%202016_pg687-700.pdf

National Research Council (U.S.). (1996). *National science education standards: Observe, interact, change, learn* (p. 3). National Academy Press.

Nggaba, M. E., Herman, T., & Prabawanto, S. (2018). Students'lateral mathematical thinking ability on trigonometric problems. *International Conference on Mathematics and Science Education of Universitas Pendidikan Indonesia*, *3*, 756–762. P-ISSN 2655-2361, E-ISSN 2655-3252.

Omran, N. J., & Krebet, D. M. (2022). Iraqi EFL preparatory students' lateral thinking and its relationship with speaking proficiency. *Nasaq Journal*, *34*(4), 526–542.

Partnership for 21st Century Skills. (2009). P21 framework definitions. Retrieved in August, 2021 from http://www.p21.org/storage/documents/P21_Framework_Definitions.pdf

Pedaste, M., Mäeots, M., Siiman, L. A., de Jong, T., van Riesen, S. A. N., Kamp, E. T., … Tsourlidaki, E. (2015). Phases of inquiry-based learning: Definitions and the inquiry cycle. *Educational Research Review*, *14*, 47–61. https://doi.org/10.1016/j.edurev.2015.02.003

Prakash, M., & Vasimalairaja, D. M. (2021). Influence of emotional intelligence and lateral thinking on achievement in biology of XI standard students. *Elementary Education Online*, *20*(5), 2426–2432.

Qi, Y. (2021). The role of mobile web platforms in the development of critical, strategic and lateral thinking skills of students in distance physical education courses. *Thinking Skills and Creativity*, *42*, 1–10. https://doi.org/10.1016/j.tsc.2021.100935

Qing, Z., Jing, G., & Yan, W. (2010). Promoting preservice teachers' critical thinking skills by inquiry-based chemical experiment. *Procedia-Social and Behavioral Sciences*, *2*(2), 4597–4603. https://doi.org/10.1016/j.sbspro.2010.03.737

Rababa, N. (2021). The effect of e-learning in developing high thinking skills. *International Journal of Data and Network Science*, *5*(1), 43–46.

Robitzsch, A. (2021). Sirt: Supplementary item response theory models. R package version 3.11-21. https://CRAN.R-project.org/package=sirt

Rosseel, Y. (2012). Lavaan: An R package for structural equation modeling. *Journal of Statistical Software*, *48*(2), 1–36. https://doi.org/10.18637/jss.v048.i02

Şahin, Ç., Arcagök, S., Çetin, Ş., Mertol, H., & Boran, G. (2017). The investigation of pre-service classroom teachers' inquiry skills. *Gazi Journal of Educational Science*, *3*(3), 17–28. https://dergipark.org.tr/tr/pub/gebd/issue/35209/390673

Saputra, H., Suhandi, A., & Setiawan, A. (2019). Profile of inquiry skills pre-service physics teacher in Aceh. International Conference on Mathematics and Science Education (ICMScE 2018). IOP Conference Series. *Journal of Physics: Conference Series*, *1157*, 032046. IOP Publishing. https://doi.org/10.1088/1742-6596/1157/3/032046, https://www.researchgate.net/publication/331663872_Profile_of_inquiry_skills_pre-service_physics_teacher_in_Aceh

Scheinin, P. M., & Mehtalainen, J. (1999). Applying the theory of knowledge to teaching thinking. In J. H. M. Hammers & J. E. H. Van Luit (Eds.), *Teaching and learning thinking skills* (pp. 81–103). Swets & Zeitlinger Publishers. https://core.ac.uk/download/93015312.pdf

Selvam, V., & Arasu, R. (2021). A review of implementing lateral thinking in higher education. *European Journal of Molecular and Clinical Medicine*, *8*(1), 2433–2437.

Semerci, Ç. (2016). Developing a lateral thinking disposition (latd) scale: A validity and reliability study. *Eğitimde Kuram ve Uygulama*, *12*(1), 358–371. https://dergipark.org.tr/tr/pub/eku/issue/26699/280892

Semerci, N. (2017). Pedagogical formation students' dispositions toward lateral thinking. *Bartin University Journal of Faculty of Education*, *6*(1), 336–345. https://doi.org/10.14686/buefad.293271

Semerci, N., Semerci, Ç., Ünal, F., Yılmaz, E., & Yılmaz, Ö. (2019). Critical thinking barriers scale: Validity and reliability studies. *Cumhuriyet International Journal of Education*, *8*(1), 281–299. https://doi.org/10.30703/cije.484271

Seranica, C., Purwoko, A. A., & Hakim, A. (2018). İnfluence of guided inquiry learning model to critical thinking skills. *IOSR Journal of Research & Method in Education (IOSR-JRME)*, *8*(1), 28–31. https://doi.org/10.9790/7388-0801022831

Şimşek, P. R., & Soytok, S. A. (2016). Comparison of inquiry skill levels of pre-service music teachers and pre-service science teachers. In *International conference on humanities and cultural studies, 6–10th November, Czech Republic* https://www.academia.edu/42908995/A_Comparison_of_Inquiry_Skill_Levels_of_Pre-service_Music_Teachers_and_Pre-service_Science_Teachers

14653435, 2024, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/ejed.12673, Wiley Online Library on [27/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Ślaski, P., & Grzelak, M. (2022). Lateral thinking in the process of logistics Students' education using the example of the EOQ model. *Polish Political Science Yearbook*, *51*, 1–9. https://doi.org/10.15804/ppsy202216

Srikongchan, W., Kaewkuekool, S., & Mejaleurn, S. (2021). Backward instructional design based learning activities to developing students' creative thinking with lateral thinking technique. *International Journal of Instruction*, *14*(2), 233–252. https://doi.org/10.29333/iji.2021.14214a

Susilawati, W., Maryono, I., Widiastuti, T., & Abdullah, R. (2018). Improvement of mathematical lateral thinking skills and student character through challenge-based learning. *Advances in Social Science, Education and Humanities Research*, *261*, 95–101. https://doi.org/10.2991/icie-18.2018.17

Tabachnick, B. G., & Fidell, L. S. (2001). *Using multivariate statistics*. Allyn & Bacon. https://faculty.chass.ncsu.edu/garson/pa765/tabachnick.htm

Tabachnick, B. G., & Fidell, L. S. (2013). *Using multivariate statistics*. Pearson.

Tornee, N., Bunterm, T., & Tang, K. N. (2017). The impacts of inquiry-based learning model on teaching science subject: A case study in Thailand. *The Turkish Online Journal of Educational Technology*, Special Issue for INTE 2017, 395–402. https://www.ic.kku.ac.th/documents/pdf/tang5.pdf

Varenina, L., Vecherinina, E., Shchedrina, E., Valiev, I., & Islamov, A. (2021). Developing critical thinking skills in a digital educational environment. *Thinking Skills and Creativity*, *41*, 100906.

Waks, S. (1997). Lateral thinking and technology education. *Journal of Science Education and Technology*, *6*(4), 245–255. https://doi.org/10.1023/A:1022534310151

Wale, B. D., & Bishaw, K. S. (2020). Effects of using inquiry-based learning on EFL students' critical thinking skills skills. *Asian-Pacific Journal of Second and Foreign Language Education*, *5*(9), 1–14. https://doi.org/10.1186/s40862-020-00090-2

Wickham, H., & Miller, E. (2021). *Haven: Import and export 'SPSS', 'Stata' and 'SAS' files*. R package version 2.4.3. https://CRAN.R-project.org/package=haven

Yazgan, A. D. (2021). Investigation of the relationship between pre-service teachers' lateral thinking levels and problem solving skills. *Journal of Theoretical Educational Science*, *14*(1), 20–37. https://doi.org/10.30831/akukeg.793247

Yıldız Feyzioğlu, E. (2015). Pre-service science teachers' pedagogical orientations of science inquiry continuum. *The Western Anatolia Journal of Educational Sciences*, *6*(11), 1–36. https://dergipark.org.tr/en/pub/baebd/issue/31824/349480

Yıldız, K., & Yılmaz, B. (2020). The relationship between critical and lateral thinking dispositions of classroom teacher candidates. *Abant İzzet Baysal University Journal of Faculty of Education*, *20*(1), 335–353. https://doi.org/10.17240/aibuefd.2020.20.52925-578949

Yılmaz, K. (2019). *Eleştirel ve analitik düşünme*. Pegem Akademi.

Yulian, R. (2021). The flipped classroom: Improving critical thinking for critical reading of EFL learners in higher education. *Studies in English Language and Education*, *8*(2), 508–522.

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14653435, 2024, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/ejed.12673, Wiley Online Library on [27/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License